Weather Analysis Results

Customer: DNV

Project Name	Project Address	Project Coordinates	
Kennedy Park Solar	United States, WI, Madison	43.0948 N	-89.2937 E

Irradiance Analysis

Irradiance Datasources

This table provides the monthly values for the currently selected dataset. To display a different dataset, choose a different dataset in the comparison table (click the column header to select), map (click a pin to select the data source), or chart (click the source name in the legend).

	NREL PSM Site	Solargis Prospect Site	NREL TMY2 MILWAUKEE	NREL TMY2 ROCKFORD	NREL TMY2 MADISON	DNV GL SunSpot Site	NREL TMY3 MADISON DA...	NREL TMY3 ROCKFORD G...	Meteonorm TMY Site
			☑	☑	☑	☑	☑	☑	
Daily GHI (kWh/m2)	4.08	3.95	3.94	3.94	3.94	3.93	3.91	3.90	3.77
Annual GHI (kWh/m2) Avg/Med: 1433/1436 StDev: 0.40%	**1490**	**1442**	**1438**	**1437**	**1436**	**1436**	**1427**	**1423**	**1375**
% from Average	4.0	0.6	0.4	0.3	0.2	0.2	-0.4	-0.7	-4.0
% from Median	3.8	0.4	0.1	0.1	0.0	0.0	-0.6	-0.9	-4.2
Distance (km)	1	0	114	101	5	6	5	101	0
Grid size (km)	4	1	-	-	-	24	-	-	-
Elevation delta (m)	-13	-1	-84	-74	-33	-22	-33	-72	-1
GHI / DIF	✓ ✓	— —	✓ ✓	✓ ✓	✓ ✓	✓ ✗	✓ ✓	✓ ✓	✓ ✓
Temp / Wind	✓ ✓	— —	✓ ✓	✓ ✓	✓ ✓	✓ ✓	✓ ✓	✓ ✓	✓ ✓
Actions	⬇	↗	⬇	⬇	⬇	⬇	⬇	⬇	⬇

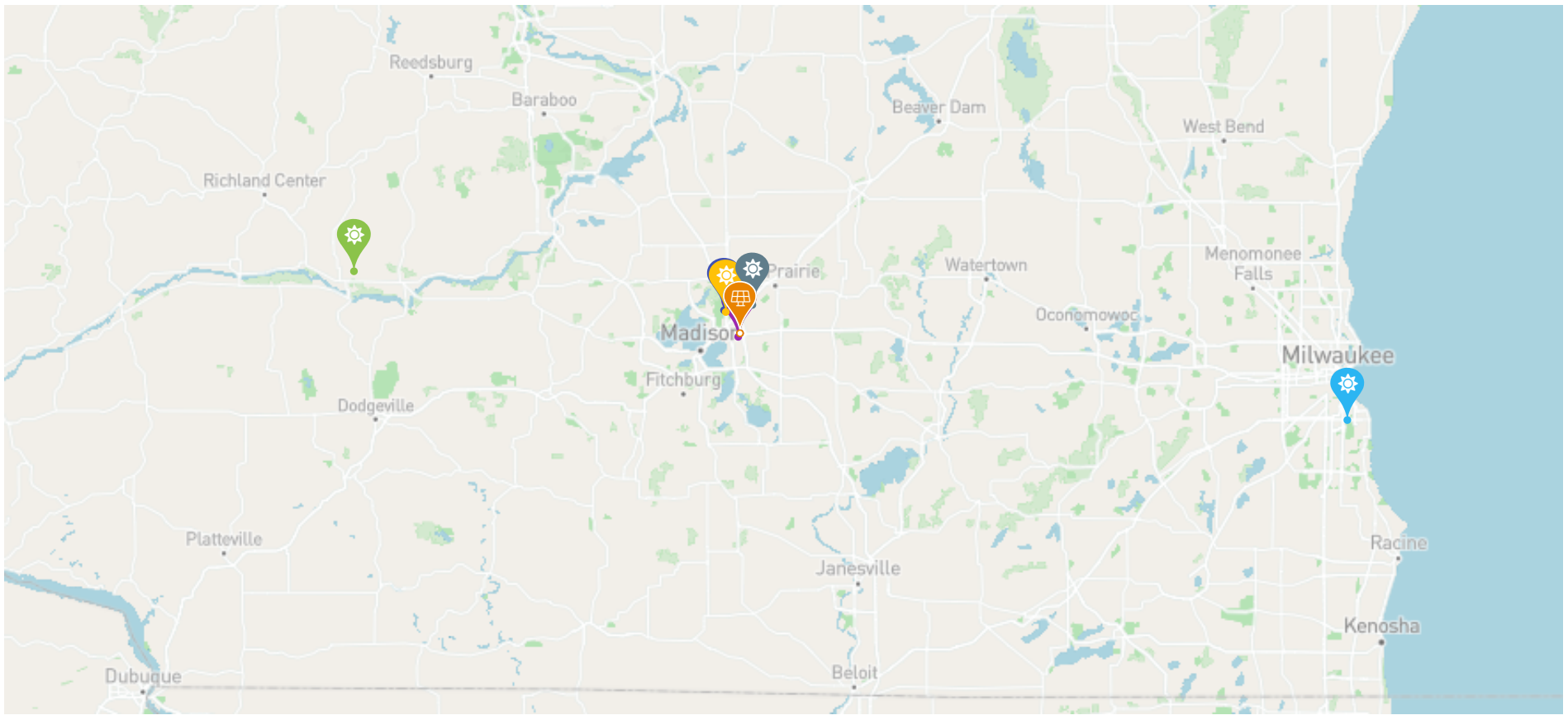
Data shown accessed on 18-Nov-2021

Daily GHI	The Daily GHI is the average daily GHI in kWh/m2 (per day). This is commonly referred to as 'sun hours,' or the equivalent number of hours at 1000 W/m2 per day. This value can be derived from the Annual GHI by dividing by 365.
Annual GHI	The Annual GHI is the annual global horizontal irradiance in kWh/m2 (per year) for each dataset.
% from Average	% from Average calculates the difference between each source and the average of the "checked" sources
% from Median	% from Median calculates the difference between each source and the median value of the "checked" sources
Distance (km)	For datasets at nearby locations, the distance reported is the approximate distance between the site location and the location of the dataset. For satellite-modeled datasets, the distance reported is the distance from the center of the applicable land grid to the project location.
Grid size (km)	Grid size is the size of the land grid to which satellite-modeled data applies.
Elevation Delta (m)	For datasets at nearby locations, the elevation delta computes the elevation difference between your site location and the location of the dataset.
GHI / DIF	GHI / DIF reports a pass/fail indicator that is based on a monthly consistency check performed by DNV. If the average monthly value for any dataset is inconsistent with the other values, it will fail the test and be indicated accordingly. The first pass/fail indicator compares GHI. The second pass/fail indicator compares diffuse irradiance. Note: Due to uncertainties in DIF data, we generally do not automatically exclude data based on failure of the DIF quality check, though we "uncheck" sources that fail the GHI quality check.
Temp / Wind	Temp / Wind reports a pass/fail indicator that is based on a monthly consistency check performed by DNV. If the average monthly value for any dataset is inconsistent with the other values, it will fail the test and be indicated accordingly. The first pass/fail indicator compares ambient temperature. The second pass/fail indicator compares wind speed. Note: Temperature and wind speed have lesser effects on energy forecasting, and we generally do not automatically exclude data based on failure of the Temp or Wind quality checks.

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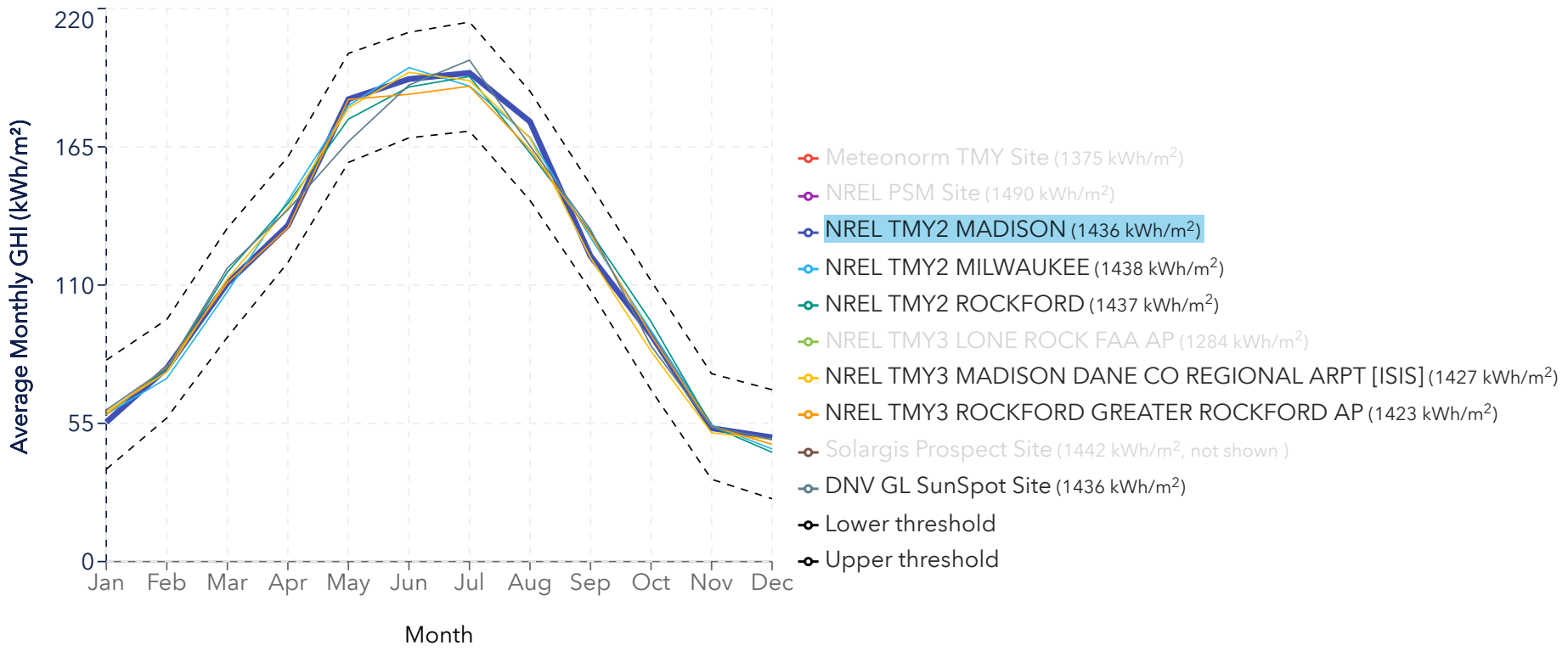
Datasource Map

This map shows the locations of your project and the locations of nearby weather stations or sources. If you upload data for locations other than your project, they will also be shown on this map.



GHI Comparison

This chart provides a visual comparison of the reported long-term average GHI for each month from each source. DNV adds the threshold lines as one indication of monthly and seasonal consistency. You can roll over each month to review the individual values. You can also click on a dataset in the legend to highlight that result in the chart.



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Summary Table

This table provides the monthly values for the currently selected dataset. You can select a new dataset by clicking the column headers above.

	NREL TMY2 MADISON			
	DHI (kWh/m²)	GHI (kWh/m²)	Temperature (°C)	Wind speed (m/s)
January	28.7	55.4	-8.5	4.5
February	34.6	76.9	-5.4	3.9
March	59.8	110.7	0.7	4.4
April	66.7	133.7	8.7	5.5
May	84.9	183.9	14.3	4.3
June	88.1	192.0	19.4	4.2
July	82.2	194.4	22.1	3.8
August	74.5	175.0	20.0	3.5
September	59.6	121.6	15.5	3.7
October	43.2	90.2	10.5	3.8
November	27.6	53.1	2.0	4.3
December	28.2	49.4	-4.5	4.9
Annual	678.1	1436.4	7.9	4.2

Soiling Snow Analysis

The Monthly Soiling and Snow Loss Table provides the monthly snow and soiling losses calculated for your project. Dust and snow losses are computed independently, and combined into a composite loss profile. The dust loss profile is generated using a modified Kimber model and the snow loss profile is generated using the Townsend model. These monthly profiles are designed for use with energy modeling tools such as PVsyst or SolarFarmer. You may adjust the inputs to match the specifics of your project, and then Save to recalculate the results. Note: Only the manual wash slider updates results in real-time as you adjust the slider.

Dust Loss Inputs

Land Use/Environment

Urban/light industrial

Number Of Manual Washes Per Year

0 1 2

Snow Loss Inputs

Mounting System

Ground mount, fixed tilt

Tilt Angle

30 degrees

Ground/Roof Clearance

3 ft / 90 cm

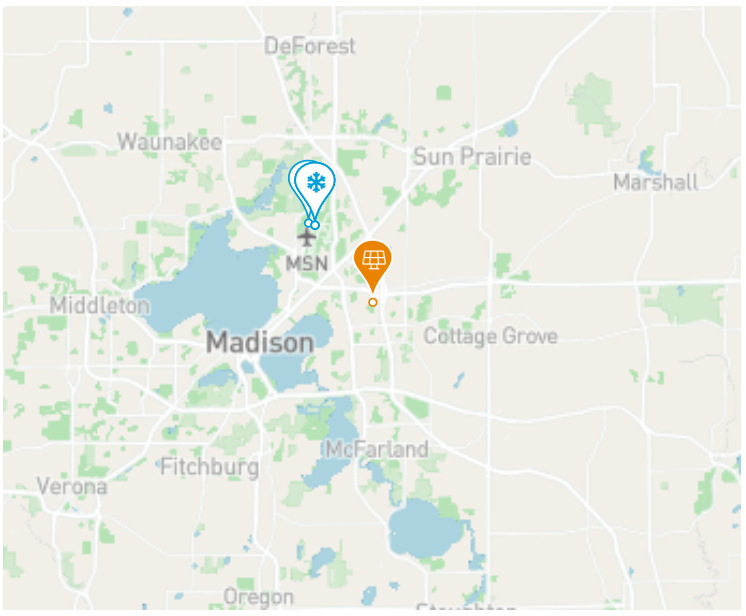
Module Configuration

3+ modules high in landscape

Module Size

Crystalline silicon (60-96 cell count, typical)

	NOAA MADISON DANE RGNL AP		NOAA MADISON DANE RGNL AP		
	Average Precipitation (inch)	Dust Soiling Loss (%)	Average Snowfall (inch)	Snow Soiling Loss (%)	Composite Loss (%)
January	1.3	3	12.9	13	13
February	1.3	3	10.6	9	9
March	2.1	1	7	5	5
April	3.2	1	2.6	2	2
May	3.5	1	0.2	0	1
June	4.3	0	0	0	0
July	3.9	1	0	0	1
August	3.9	1	0	0	1
September	3.2	1	0	0	1
October	2.4	1	0.5	0	1
November	2.1	1	3.6	3	3
December	1.6	1	13.5	10	10



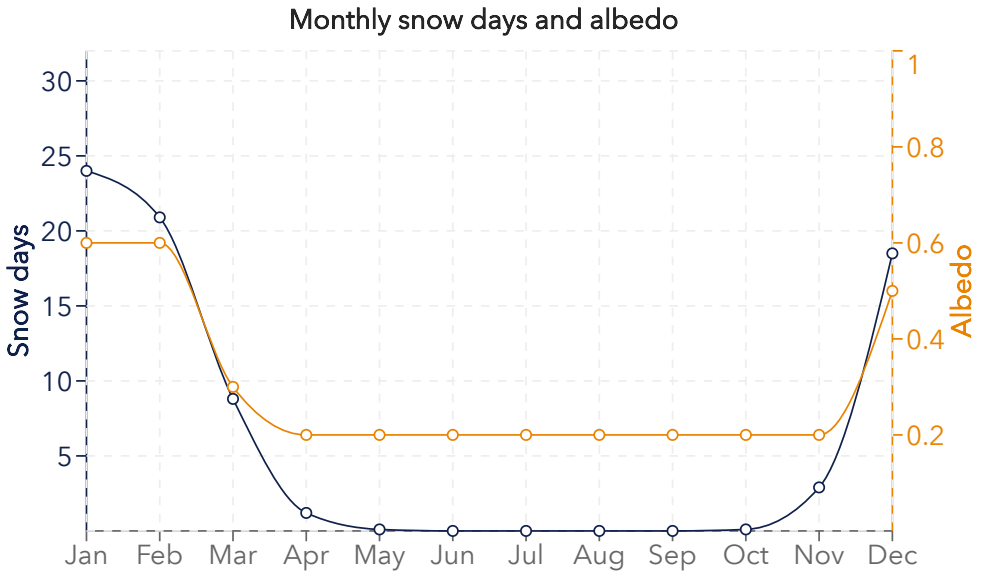
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Albedo Analysis BETA

The Monthly Albedo Profile provides the monthly albedo calculated for the selected project location. Albedo is a measure of the amount of irradiance reflected off of the ground or roof, and is presented as a fraction between 0 and 1. An albedo of 0.2 means that 20% of the irradiance is reflected off the ground.

The albedo value impacts the estimated plane of array (POA) irradiance on the front of the modules since a portion of POA irradiance is attributed to irradiance reflected off of the ground. For solar projects that use bifacial modules, this value also determines how much of the irradiance is available to the backside of the modules. During months with no snow, the albedo is driven primarily by the type of ground cover (e.g. sand, grass, dirt) and precipitation profile (wet ground has a lower albedo). Most natural ground types will have an albedo of between 0.15 and 0.25, with some monthly variation expected due to precipitation. During winter months, the albedo is driven primarily by the average number of days with snow cover on the ground. Fresh snow has a very high albedo (0.8), so months with consistent snow cover will have a higher average albedo.

		NOAA MADISON DANE RGNL AP	
	Average Snowfall (inch)	Average Days with Snow Cover	Albedo
January	12.9	24	0.6
February	10.6	20.9	0.6
March	7	8.8	0.3
April	2.6	1.2	0.2
May	0.2	0.1	0.2
June	0	0	0.2
July	0	0	0.2
August	0	0	0.2
September	0	0	0.2
October	0.5	0.1	0.2
November	3.6	2.9	0.2
December	13.5	18.5	0.5



In the table above, DNV has applied an albedo for most ground types of 0.2 (for periods without snow cover). This value is appropriate for most common vegetation and soil types. DNV notes the following considerations in assessing albedo values for projects with other surface types:

- Examples of surface types with albedo lower than 0.15 include wet earth, asphalt, and water.
- Examples of surface types with albedo higher than 0.25 include light/white sand, concrete or roof membranes.
- For projects with these surface types, it is recommended to review the surface type with a DNV engineer and/or collect on-site measurements to aid in determining the most appropriate albedo value.
- High albedo surfaces (e.g. white roof membranes, fresh concrete) often do not maintain a high albedo without regular cleaning. To assess the potential impact of albedo loss over time due to soiling, consult with a DNV engineer.

Note: Solar Resource Compass by DNV is intended to be used for accurate, indicative estimates of solar irradiance, soiling and other site characteristics. Solar Resource Compass utilizes the same analytics engine employed by the DNV solar energy analytics team. Results from Solar Resource Compass will closely match results prepared manually by DNV's analysts as part of a project financing review. In some cases, differences due to microclimatic conditions can be determined during a manual review by a DNV engineer. For questions or support, please contact your Project Manager or e-mail us at src_support@dnv.com.

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